Temperature-Driven Power Demand in Distributed Sensor Sites: A Regression and Control-Oriented Analysis

Muhammad Uzair Javed

***Abstract*—We study how ambient temperature drives electrical power draw across three monitoring sites using 2880 hourly observations. A piecewise-linear demand model is fitted by least squares, its estimator variance is derived, and a closed-loop controller is proposed to regulate demand. The analysis is a typesetting and reproducibility demonstration: the dataset is synthetic and every figure and table is regenerated by script.**

***Index Terms*—power demand, regression, PID control, reproducible research, LaTeX.**

I. Introduction

RELIABLE demand models are central to energy monitoring. We combine data handling in `pandas` [1], visualization with Matplotlib and seaborn [2], [3], and classical control analysis [4]. Colour choices follow a colour-blind-safe palette [5]. Section II defines the model, Section III the controller, and Section IV the results.

II. Demand Model

Let T_k denote temperature and P_k power at hour k. We assume a hinge model with threshold τ:

$$P_k = \beta_0 + \beta_1 \max(T_k - \tau,\, 0) + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

Stacking observations gives $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{X} \in \mathbb{R}^{N\times 2}$. The least-squares estimator and its covariance are

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y}, \tag{2a}$$

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}. \tag{2b}$$

For the two-parameter design, with $h_k = \max(T_k - \tau, 0)$, the normal equations read

$$\begin{bmatrix} N & \sum h_k \\ \sum h_k & \sum h_k^2 \end{bmatrix} \begin{bmatrix} \hat\beta_0 \\ \hat\beta_1 \end{bmatrix} = \begin{bmatrix} \sum P_k \\ \sum h_k P_k \end{bmatrix}. \tag{3}$$

The residual variance is estimated by

$$\hat\sigma^2 = \frac{1}{N-2}\sum_{k=1}^{N} \big(P_k - \hat\beta_0 - \hat\beta_1 h_k\big)^2,$$

$$\operatorname{Var}(\hat\beta_1) = \frac{\sigma^2}{\sum_k (h_k - \bar h)^2}. \tag{4}$$

Independent researcher. This paper is a portfolio typesetting demonstration using synthetic data. Source code: https://github.com/uzairjaved-prog/Academic-typesetting-portfolio.

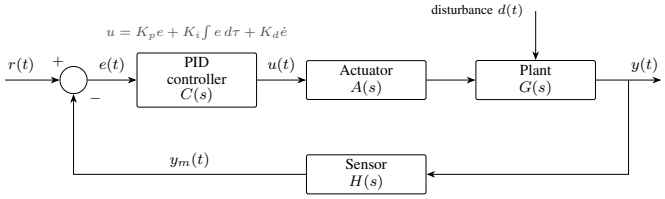


Fig. 1: Closed-loop PID control of the power plant (native TikZ).

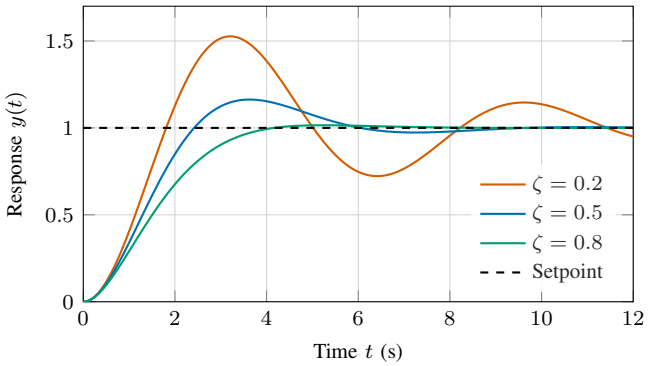


Fig. 2: Unit-step responses of (7) for three damping ratios (PGFPlots).

Properties used later are collected in

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \qquad \operatorname{tr}\big[\operatorname{Cov}(\hat{\boldsymbol{\beta}})\big] = \sigma^2 \operatorname{tr}\big[(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\big], \tag{5}$$

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \succeq (\boldsymbol{X}^\top \boldsymbol{X}/\sigma^2)^{-1} \quad \text{(Cramér–Rao [6])}.$$

III. Control-Oriented Formulation

To regulate demand we close a feedback loop (Figure 1) with PID law

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d\, \dot e(t). \tag{6}$$

The plant is modelled as a second-order system with natural frequency ω_n and damping ζ [4]:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \tag{7}$$

Figure 2 shows step responses; light damping produces the overshoot that the derivative term suppresses. A supervisory finite-state logic [7] switches the controller mode (Figure 3).

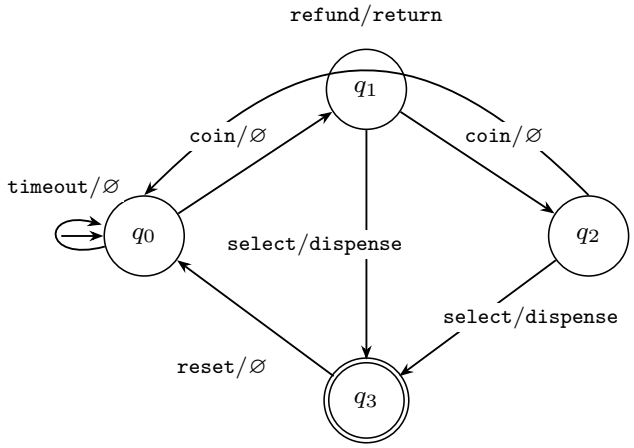


Fig. 3: Supervisory finite-state machine for mode switching.

TABLE I: Per-site summary statistics (synthetic data).

Site	n	$\bar{T}$ (°C)	$\bar{P}$ (W)	s_P (W)	$\bar{a}$ (g)
Site A	1151	19.87	133.74	16.60	0.377
Site B	989	20.18	139.18	16.92	0.385
Site C	740	19.87	131.00	16.21	0.383

IV. Results

Across 2880 observations the Pearson correlation between temperature and power is $r = 0.86$ (Figure 4(b)). Table I lists per-site statistics computed by `generate_figures.py`; Site B draws the most power on average, consistent with the offsets visible in Figure 4(c).

The hinge model (1) explains the knee visible in Figure 4d. Fits for (2a) converge in one step because the problem is linear in $\boldsymbol{\beta}$ once τ is fixed.

V. Conclusion

We demonstrated a reproducible pipeline from raw CSV to camera-ready PDF: scripted figures, native TikZ diagrams, `booktabs`/`siunitx` tables and BibTeX references.

References

[1] W. McKinney, "Data structures for statistical computing in Python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 56–61.

[2] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

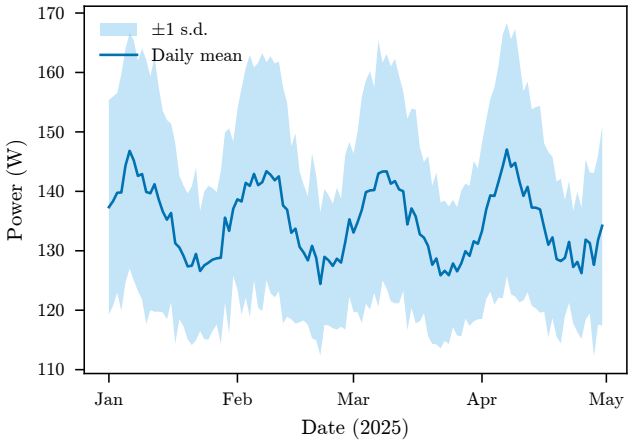
[3] M. L. Waskom, "seaborn: statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021.

[4] K. Ogata, *Modern Control Engineering*, 5th ed. Upper Saddle River, NJ: Prentice Hall, 2010.

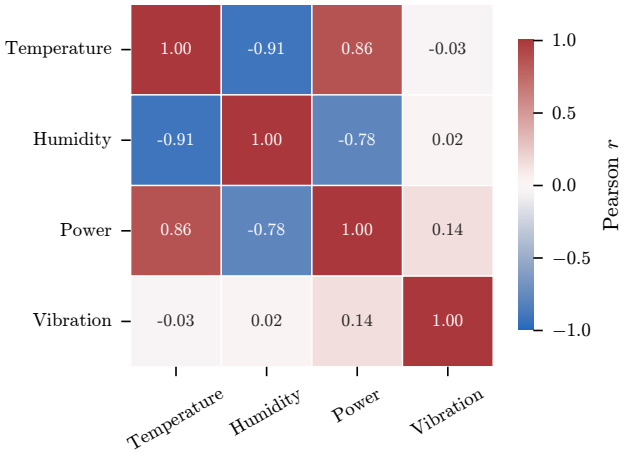
[5] B. Wong, "Points of view: Color blindness," *Nature Methods*, vol. 8, no. 6, p. 441, 2011.

[6] S. M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*. Englewood Cliffs, NJ: Prentice Hall, 1993.

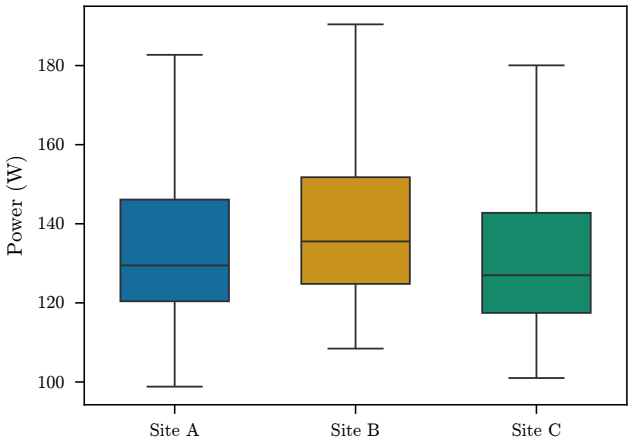
[7] J. E. Hopcroft, R. Motwani, and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, 3rd ed. Boston, MA: Addison-Wesley, 2006.



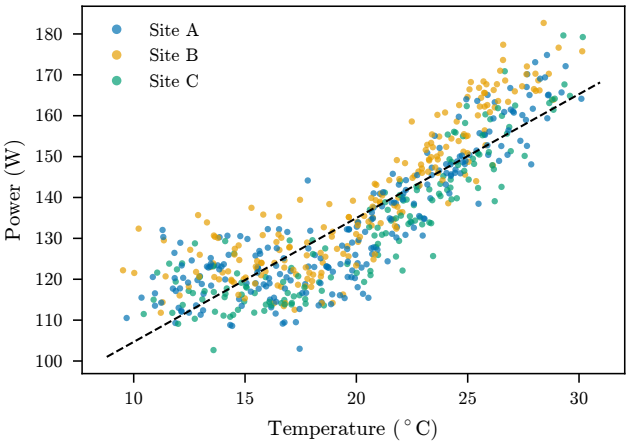
(a) Daily mean power with ±1 s.d. band.



(b) Correlation heatmap.



(c) Power distribution by site.



(d) Power versus temperature with linear fit.

Fig. 4: Exploratory analysis of the sensor dataset. Panels are separate vector PDFs exported from Matplotlib in Computer Modern.